

Grade 8
Unit 4
Lesson 5 Practice

Name _____
Date _____
Period _____

Hadley is evaluating the expression $\frac{(4.5)^{-5} \cdot (4.5)^0}{(4.5^3)^{-2}}$.

1. Which of the expression(s) show the correct first step in evaluating the expression above?

A. $\frac{(4.5)^{-5}}{(4.5^3)^{-2}}$

B. $\frac{(4.5)^0}{(4.5)^{-6}}$

C. $\frac{(4.5)^{-5} \cdot (4.5)^0}{(4.5)^{-6}}$

D. $\frac{(4.5)^{-5} \cdot (4.5)^0}{(4.5)^6}$

2. Which exponent law(s) was applied in the first step?

- ☐ Zero Exponent
- ☐ Identify Exponent
- ☐ Negative Exponent
- ☐ Power of a Power
- ☐ Product of Powers
- ☐ Power of a Product
- ☐ Quotient of Powers
- ☐ Power of a Quotient

3. Evaluate the expression $\frac{(4.5)^{-5} \cdot (4.5)^0}{(4.5^3)^{-2}}$.

Evaluate the numerical expressions. Show all the steps using the laws of exponents.

4. $\left(\frac{3^4}{3^{-2}}\right)^{-3}$

5. Which exponent law was applied first when evaluating question 4?

6. $\frac{(3 \cdot 8)^3}{3^2 \cdot 3^2}$

7. Which exponent law was applied first when evaluating question 6?

8. $\frac{\left(\frac{2}{3}\right)^{-5} \cdot \left(\frac{2}{3}\right)^0}{\left(\frac{2}{3}\right)^{-2}}$

9. Which exponent law was applied first when evaluating question 8?

10. Lauren evaluated the expression $\frac{(6^{-1}) \cdot (6^5)^0}{6^2}$. Her work is shown below. Circle the step where she made an error and describe the mistake she made.

Problem	Describe the Mistake
$\frac{(6^{-1}) \cdot (6^5)^0}{6^2}$	
$\frac{(6^{-1}) \cdot 6^5}{6^2}$	
$\frac{6^4}{6^2}$	
6^3	
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